Time Series Analysis

Mock test for Practicals

- **For the entire test $\{Z_t\} \sim \mathrm{WN}(0, 1)$.**
- **Explain all your answers.**
- **You can refer to any result obtained in class.**
- **Double answers, one correct one false $\Longrightarrow$ both false.**
- **Write in a readable way.**
- **Number of points in brackets [..]**

1) Let

$$X_t = 1 + Z_1 Z_t, \quad t \geq 2.$$

Show why, or why not, $\{X_t\}$ is a stationary process.

2) Let

$$p(z) = 1 - \frac{1}{2}z + z^2.$$

Is the process $\{X_t\}$ defined via the equation $p(B)X_t = Z_t$ a stationary and causal process? Explain your answer.

3) Consider the MA(2) process $X_t = Z_t + Z_{t-1}/2 + Z_{t-2}/4$. Derive the long-run variance of $\{X_t\}$.

4) We have observed time series data from the process in Exercise 3. Based on $x_{20} = 2$ and $x_{19} = 1$ give the optimal linear forecast for x_{21}.

5) Consider the model

$$X_t + \theta X_{t-1} = Z_t + Z_{t-1}/4.$$

What type of model is this? Discuss whether there exists a stationary, causal and invertible solution.